

Artificial Intelligence Using Python

Hands-on Exercise No. 1

By Bakht Zamin

Problems Solution

Task 1:

Write a program that takes salary as input. Using conditional statements, calculate the final tax based on the following rules:

- If the salary is less than 30,000 → Tax rate is 5%
- If the salary is between 30,000 and 70,000 → Tax rate is 15%
- If the salary is greater than 70,000 → Tax rate is 25%.

Solution:

```
1 # Take input from user
salary = float(input("Enter your salary: "))

# Apply conditions
if salary < 30000:
    tax_rate = 0.05
elif salary <= 70000:
    tax_rate = 0.15
else:
    tax_rate = 0.25

# Calculate tax
tax = salary * tax_rate

# Display result
print("Salary:", salary)
print("Tax Rate:", tax_rate * 100, "%")
print("Final Tax:", tax)

... Enter your salary: 70000
Salary: 70000.0
Tax Rate: 15.0 %
Final Tax: 10500.0
```

Task 2:

Given a list of words: words = ["apple", "banana", "kiwi", "cherry", "mango"]

Create a dictionary that maps each word to its corresponding length.

Example Output:

{'apple': 5, 'banana': 6, 'kiwi': 4, 'cherry': 6, 'mango': 5}

```

words = ["apple", "banana", "kiwi", "cherry", "mango"]

word_length = {}

for word in words:
    word_length[word] = len(word)

print(word_length)

... {'apple': 5, 'banana': 6, 'kiwi': 4, 'cherry': 6, 'mango': 5}

```

Task 3:

Create a Chat System using Object-Oriented Programming (OOP) concepts.

We need to create the following classes:

- User
- Message
- ChatRoom

The system should implement the following functionalities:

- Sending messages
- Viewing chat history
- User joining and leaving the chat room

Solution:

```

('apple': 5, 'banana': 6, 'kiwi': 4, 'cherry': 6, 'mango': 5)

11 class User:
12     def __init__(self, name):
13         self.name = name
14
15     class Message:
16         def __init__(self, sender, text):
17             self.sender = sender
18             self.text = text
19
20     class ChatRoom:
21         def __init__(self):
22             self.users = []
23             self.messages = []
24
25         def join(self, user):
26             self.users.append(user)
27             print(user.name, "joined the chat")
28
29         def leave(self, user):
30             self.users.remove(user)
31             print(user.name, "left the chat")
32
33         def send_message(self, user, text):
34             message = Message(user, text)
35             self.messages.append(message)
36
37         def show_chat(self):
38             for msg in self.messages:
39                 print(msg.sender.name + ": ", msg.text)
40
41 # Example Usage
42 u1 = User("Ali")
43 u2 = User("Ahmed")
44
45 chat = ChatRoom()
46
47 chat.join(u1)
48 chat.join(u2)
49
50 chat.send_message(u1, "Hello")
51 chat.send_message(u2, "Hi")
52
53 chat.show_chat()
54
55 chat.leave(u1)
56
57 ... Ali joined the chat
58 ... Ahmed joined the chat
59 ... Ali: Hello
60 ... Ahmed: Hi
61 ... Ali left the chat
62
63 11 Start coding or generate with AI.

```